Project Proposal

End-to-End Secure Real-Time Chat Application

Course: Secure Software Design

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1 Abstract

This project proposes the design and development of an end-to-end secure real-time chat application with a strong emphasis on secure software design principles. The system integrates modern web technologies, real-time communication, and proactive vulnerability assessment using automated scanning tools. The primary goal is to mitigate common web application vulnerabilities and demonstrate secure development and deployment practices suitable for enterprise environments.

2 Introduction

Real-time communication systems are increasingly targeted by cyber attacks due to the sensitive nature of transmitted data. This project focuses on building a secure chat application using a security-first approach. The application is designed to prevent common vulnerabilities, ensure confidentiality and integrity of data, and provide a controlled environment for attack simulation and security testing.

3 Project Objectives

- Design a real-time chat application with strong security controls.
- Implement secure authentication and authorization mechanisms.
- Mitigate OWASP Top 10 web application vulnerabilities.
- Perform vulnerability scanning and attack simulation using Nessus.
- Apply DevSecOps practices using containerization and CI/CD pipelines.

4 Scope of the Project

The project scope includes secure real-time messaging, user authentication, role-based access control, encrypted data storage, audit logging, and automated vulnerability assessment. Nessus will be used to simulate real-world attacks and identify security weaknesses. The system is intended for internal enterprise communication and cybersecurity learning purposes.

5 System Architecture

The proposed system follows a layered architecture consisting of a web-based frontend, a FastAPI backend, a real-time communication layer, a secure database, and a security monitoring component.

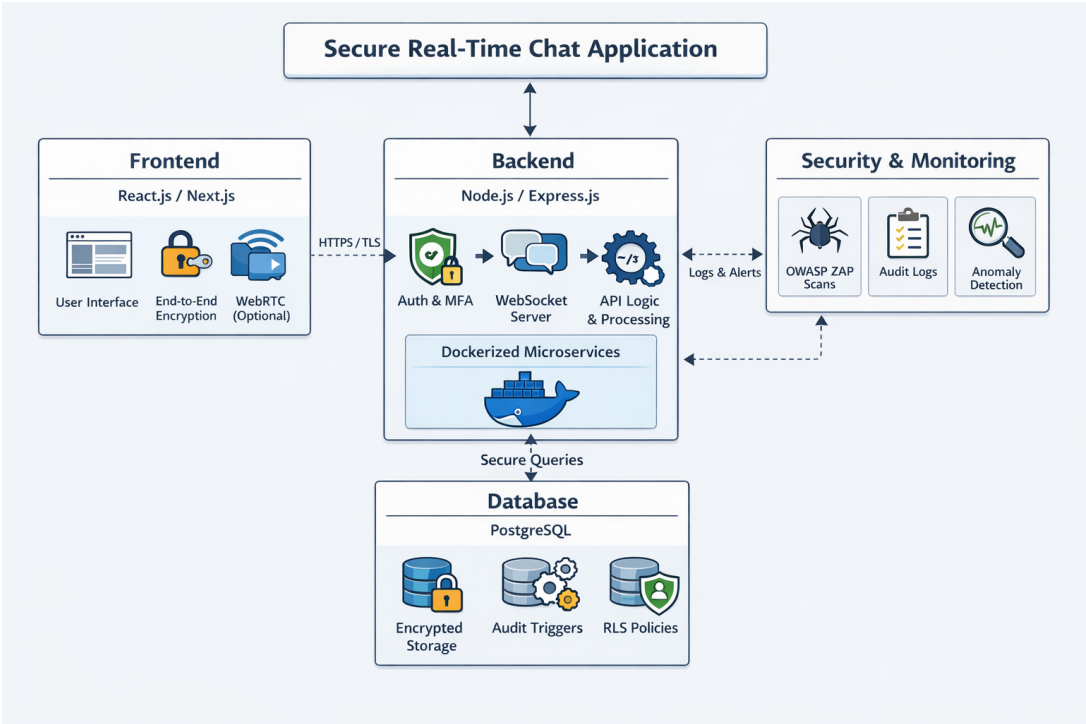

Figure 1: Proposed Secure Real-Time Chat Application Architecture

The architecture diagram illustrates interaction between the frontend, FastAPI backend, WebSocket server, PostgreSQL database, CI/CD pipeline, and vulnerability scanning tools.

6 Technologies Used

- Frontend: Next.js
- Backend: FastAPI (Python)
- Real-Time Communication: WebSockets
- Database: PostgreSQL
- Containerization: Docker
- CI/CD: GitHub Actions
- Security Testing: Nessus, OWASP ZAP

7 Security Testing and Attack Simulation

Nessus will be used to perform vulnerability scanning and simulate real-world attacks on the deployed application. Identified vulnerabilities will be analyzed and mitigated using secure coding practices. This approach helps validate the effectiveness of implemented security controls and provides hands-on experience in offensive and defensive security techniques.

8 Conclusion

This project bridges theoretical secure software design concepts with practical implementation. By integrating FastAPI, real-time communication, and vulnerability assessment tools such as Nessus, the system demonstrates industry-relevant security practices and provides significant learning value for cybersecurity students.